Total No. of Questions : 8 | | Total No. of Printed Pages : 3

Paper Code : 21308

F-408

B.C.A. (Second Semester)

Examination, 2023

(New Course)

Paper No. BCA-N-203

MATHEMATICS-II

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used are as usual.

1. (a) Show that every infinite bounded set has at least one limitting point.

   (b) The intersection of a finite collection of open sets in R is an open set in R.

2. (a) Show that $\lim_{(x,y)\to(-1,2)} \frac{x^3+y^3}{x^2+y^2} = \frac{7}{5}$

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   (b) Find the maximum value of $(ax + by + cz)e^{-\alpha^2x^2-\beta^2y^2-\gamma^2z^2}$

3. (a) Prove that $\lim_{n\to\infty} \frac{1}{n}[(n+1)(n+2)\ldots(2n)]^{1/n} = \frac{4}{e}$

   (b) Test for the convergence or divergence of the series Σu_n where $u_n = \sqrt{(n^4+1)} - \sqrt{(n^4-1)}$

4. (a) Test for convergence the series

   $x \log x + x^2 \log(2x) + x^3 \log(3x) + \ldots\ldots\ldots + x^n \log(nx) + \ldots\ldots$

   (b) Show that a monotonic increasing sequence is convergent iff it is bounded.

5. (a) Expand log sec x by Maclaurin's theorem.

   (b) If $\phi(x) = \sqrt{x}, f(x) = \frac{1}{\sqrt{x}}$, then show that c is the geometric mean of a, b in Cauchy's M.V.T. while c is the harmonic mean of a, b if $\phi(x) = \frac{1}{x^2}, f(x) = \frac{1}{x}$

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6. State and prove comparison test.

7. (a) Show that the function

$$f(x) = \begin{cases} \dfrac{xe^{1/x}}{(1+e^{1/x})} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$ is continuous

at x=0 but not differentiable at x=0.

(b) Evaluate $\lim_{x \to 0} \left(\frac{\tan x}{x}\right)^{1/x}$

8. (a) Show that if the perimeter of a triangle is constant, its area is a maximum when it is equilateral.

(b) Show that the sequence $\left\langle \frac{2^n}{n!} \right\rangle$ is a monotonic decreasing sequence. Also show that it is bounded.

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